

EXECUTIVE OVERVIEW & TECHNICAL ARCHITECTURE

ENX MONEY

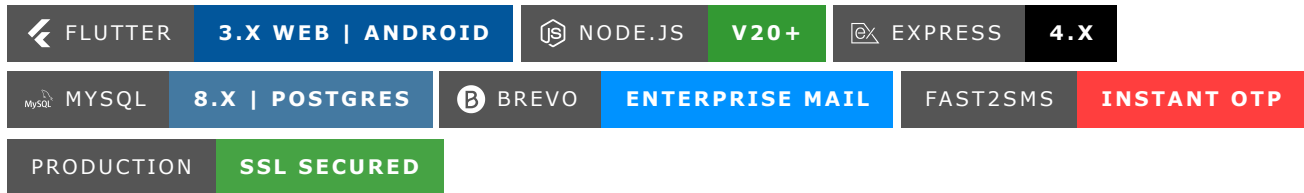
Platform Overview, Infrastructure & Compliance Specification

Next-Gen Minimalist Fintech Application for High-Net-Worth Wealth, Liabilities, Expenses & Smart Khata

Organization:	Enterprenex Solutions Pvt. Ltd.
Official Web App:	https://enxmone.yp.terprenex.solutions
Corporate Portal:	https://enterprenex.solutions
Official Support:	support@enterprenex.solutions
Repository:	Enterprenex-Solution-Pvt-Ltd/Enx-Money
Release Version:	4.2.0 (Production Hardened Release)
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ENX Money — Full-Stack Fintech & Smart Wealth Platform

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Institutional-Grade Fintech Application for High-Net-Worth Wealth, Liabilities, Loan EMI Management, Expense Analytics & Smart Business Khata

Inspired by the high-contrast luxury dark aesthetic of CRED and the autonomous financial intelligence of Walnut.

[🌐 Live Production Web App](#) • [🏢 Corporate Portal](#) • [📱 Download Android APK](#)

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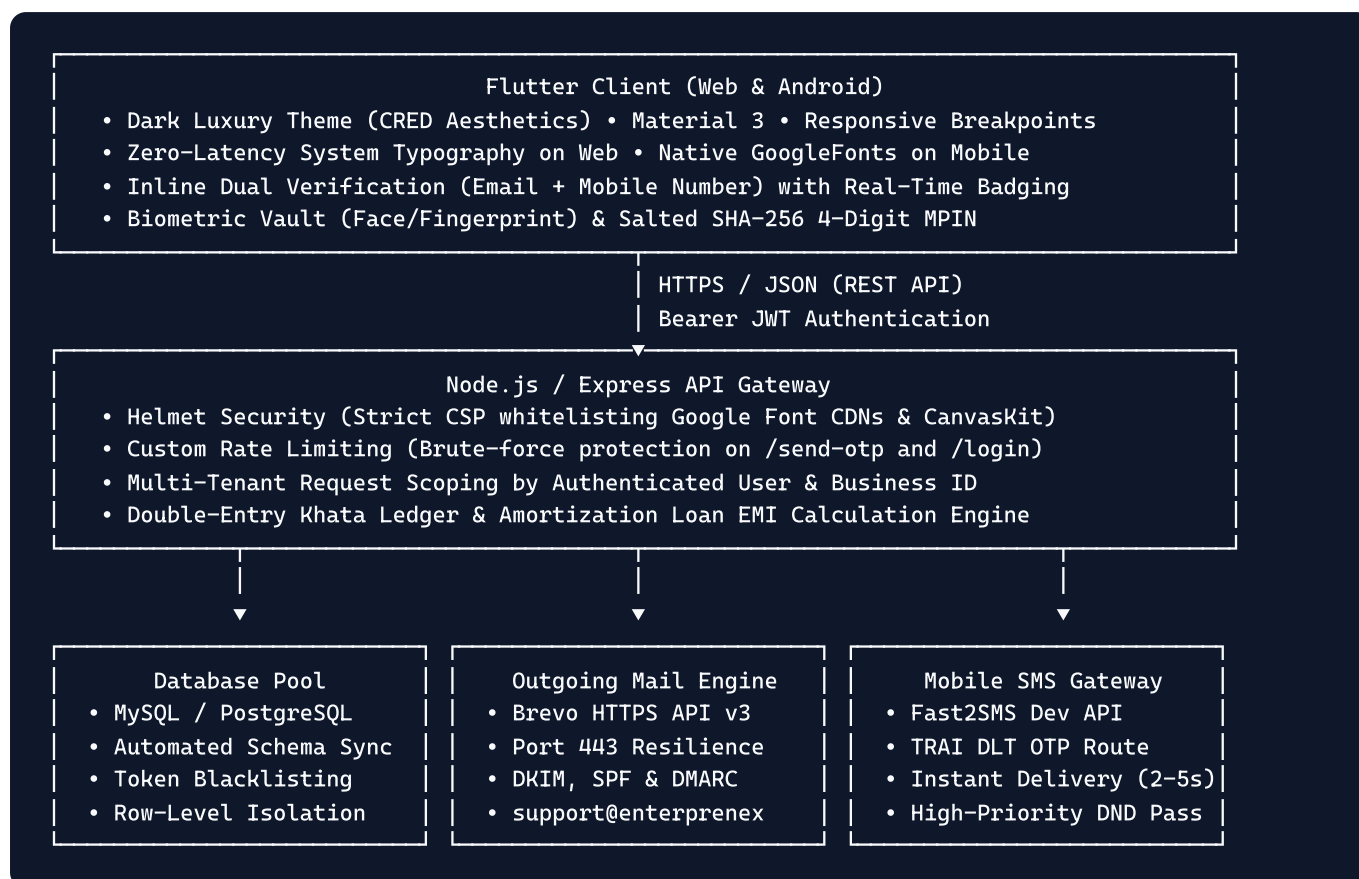
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Live Production Deployments & URLs

The entire ENX Money ecosystem is deployed on enterprise cloud infrastructure with continuous deployment from GitHub:

Service / Component	Live Production URL	Description
Official Web Application	https://enxmoney.enterprenex.solutions	Full Flutter Web SPA matching mobile UI, optimized with zero-latency typography & responsive fintech desktop framing.
Corporate Apex Domain	https://enterprenex.solutions	Official enterprise gateway and public domain.
Production REST API	https://enxmoney.enterprenex.solutions/api/v1	Secure Node.js/Express API with Rate Limiting, Helmet, and JWT authorization.
API Healthcheck	https://enxmoney.enterprenex.solutions/health	Real-time service uptime, database connectivity, and environment status.
Direct APK Download	https://enxmoney.enterprenex.solutions/download-apk	Direct Android production APK binary distribution portal.
Legal Privacy Policy	https://enxmoney.enterprenex.solutions/privacy-policy	Public statutory policy compliant with Indian DPDP Act 2023 & Google Play rules.
Account Deletion Portal	https://enxmoney.enterprenex.solutions/delete-account	User-facing self-service account and data deletion request portal.

System Architecture



Enterprise Communications & Email Infrastructure

The platform uses a dedicated, authenticated corporate email and forwarding architecture:

1. Outgoing Transactional Email (Brevo HTTPS API)

- **Official Sender:** ENX Money <support@enterprenex.solutions>
- **Delivery Protocol:** Native HTTPS API (POST `https://api.brevo.com/v3/smtp/email`) over port 443 — immune to cloud provider SMTP port blocking (ports 25, 465, 587).
- **Domain Authentication (100% Verified in DNS):**
 - **Brevo Code:** `brevo-code:93fc028e2562587764eaf2c03c0b45fd` (GoDaddy TXT Record)
 - **DKIM 1 & DKIM 2:** 2048-bit domain keys ensuring tamper-proof cryptographic delivery.
 - **DMARC Record:** Strict quarantine enforcement (`v=DMARC1; p=quarantine;`).
 - **Combined SPF Record:** `v=spf1 include:spf.improvmx.com include:_spf.brevo.com ~all` .

2. Inbound Customer Support Routing (ImprovMX)

- **Catch-All Wildcard Rule:** `*@enterprenex.solutions` $\rightarrow$ Forwarded instantly to corporate inbox `enxproductofficial@gmail.com`.
- **Active Operational Aliases:**
 - `support@enterprenex.solutions` — Customer support & ticket resolution
 - `info@enterprenex.solutions` — General enquiries & public information
 - `contact@enterprenex.solutions` — Business partnerships & corporate communication
 - `sales@enterprenex.solutions` — Subscriptions & enterprise onboarding
 - `billing@enterprenex.solutions` — Payment confirmations & billing inquiries
 - `admin@enterprenex.solutions` — Systems & infrastructure governance
 - `feedback@enterprenex.solutions` — Customer suggestions & feedback
 - `help@enterprenex.solutions` — General platform assistance

Authentication, Biometrics & Dual OTP Verification

1. Registration Flow with Inline Verification

The signup form (`CreateAccountScreen`) features independent verification directly within the form fields:

- **Email Verification:** Interactive `[ Verify ]` button inside the email input field triggers an OTP dispatch via Brevo. Entering the 6-digit code in the modal transforms the button into a green `[ ✓ Verified ]` badge.
- **Mobile Number Verification:** Interactive `[ Verify ]` button inside the mobile input field triggers an instant SMS OTP via Fast2SMS. Entering the 6-digit code marks the phone number as `[ ✓ Verified ]`.
- **Account Creation:** Submitting the form creates the account and binds both verified channels directly to the user's primary database profile.

2. Returning User Login Options

The login screen (`LoginScreen`) provides 3 streamlined authentication options:

1. **Option A: Email + Password** — Traditional credential login with Bcrypt password hashing.
2. **Option B: Mobile / Email + OTP** — Passwordless sign-in for users entering either their registered phone number or email address.
3. **Option C: Quick Unlock (Biometric & 4-Digit MPIN)** — Returning users on recognized devices unlock the app in <1 second using device biometrics (Fingerprint / Face ID) or a salted SHA-256 MPIN, eliminating repetitive OTP costs.

Flutter Web & Mobile Responsive Architecture

1. Zero-Latency Typography Engine on Web

- Replaced runtime font downloads with an instant native typography fallback stack on web (`Segoe UI` , `Roboto` , `San Francisco` , `Arial`).
- Mobile Android builds continue leveraging luxury `GoogleFonts.outfit` and `GoogleFonts.spaceGrotesk` .
- Configured Helmet Content Security Policy (`connectSrc` , `fontSrc` , `styleSrc`) whitelisting Google Font CDNs and CanvasKit.

2. Centered Responsive Desktop Card Framing

- On ultra-wide monitors (1080p, 1440p, 4K), forms and onboarding flows are wrapped in centered `ConstrainedBox(maxWidth: 500)` containers.
 - Forms render as sleek, elegant fintech cards rather than stretching 1920px wide across desktop viewports.
-

Production Release Artifacts & Play Store Compliance

1. Android Binaries

- **Google Play App Bundle (AAB):** `ENX-Money.aab` (~58.4 MB)
Built with Gradle 8, Android NDK 28, and minified R8 code shrinking for Play Store console submission.
- **Standalone Android APK:** `ENX-Money.apk` (~59.4 MB)
Direct APK binary with multi-architecture support (armeabi-v7a, arm64-v8a, x86_64).

2. Google Play Store Compliance (15 Policy Areas)

Fully documented in `docs/PROJECT_COMPLIANCE.md` :

- **Official Developer Support Email:** `support@enterprenex.solutions`
 - **Official Support Phone:** `+91 9226860060`
 - **Official Corporate Website:** `https://enxmoney.enterprenex.solutions`
 - **Public Privacy Policy:** `https://enxmoney.enterprenex.solutions/privacy-policy`
 - **Account Deletion URL:** `https://enxmoney.enterprenex.solutions/delete-account`
 - **Terms & Conditions:** `https://enxmoney.enterprenex.solutions/terms`
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CI/CD & Automated Verification

1. Running Backend Test Suites

```
cd server
npm test
```

Executes all 14 Jest test suites validating Auth, Cards, Invoices, Khata Ledger, KYC, and Razorpay standard checkout.

2. Testing End-to-End OTP Dispatches

```
# Verify Email OTP via Brevo HTTPS API:
node -e "require('./src/services/email.service').sendOtpEmail({ email: 'support@enterprenex.solution' })"

# Verify Fast2SMS Mobile OTP:
node -e "require('./src/services/sms.service').sendOtpSms({ phone: '+919440829762', otp: '123456' })"
```

3. Compiling Production Web App

```
cd client
flutter build web --release
# Sync built output to backend public web directory:
Copy-Item -Path "client\build\web\*" -Destination "server\public\" -Recurse -Force
```

Repository Structure

```

ENX_Money/
├── client/                                # Production Flutter Multi-Platform App (Web + Android)
│   ├── lib/
│   │   ├── core/                        # Design tokens, typography fallback, network clients
│   │   └── features/                    # Auth, Cards, Khata, Expenses, Loans, Profile, KYC
│   └── web/                             # Web configuration, manifest.json, index.html
├── server/                              # Production Node.js / Express API Backend
│   ├── public/                         # Production Flutter Web distribution bundle (served at /)
│   ├── src/
│   │   ├── config/                     # Environment, DB connection pool, Brevo mailer, SMS
│   │   ├── controllers/                # Request handlers (Auth, KYC, Expenses, Cards, UPI)
│   │   ├── models/                    # Database access models & tenant isolation
│   │   ├── routes/                    # REST API route declarations
│   │   └── services/                  # Brevo EmailService, Fast2SMS SmsService, TokenService
│   └── tests/                          # 14 Jest automated test suites
├── docs/                               # Google Play policies, compliance docs & data safety
├── render.yaml                         # Infrastructure-as-code deployment configuration
├── ENX-Money.aab                       # Google Play Console production release bundle
├── ENX-Money.apk                       # Direct install Android release APK
└── DOCUMENTATION.md                    # Comprehensive 64-chapter engineering manual

```

<div align="center"> ENX Money • Engineered with Precision by Enterprenex Solutions Pvt. Ltd.

<i>Official Support: support@enterprenex.solutions </i>

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